HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG

KHOA CÔNG NGHỆ THÔNG TIN I

KIẾN TRÚC VÀ THIẾT KẾ PHẦN MỀM

Giảng viên hướng dẫn : Trần Đình Quế

Lớp : D22CNPM03

Họ và tên : Phan Văn Biên

Mã sinh viên : B22DCCN078

Hà Nội – 2025

I. Màn hình tạo project và server

1. Khởi tạo mã nguồn Django

```
mkdir laptop_service mobile_service staff_service customer_service
```

```
cd laptop_service
django-admin startproject laptop_service .
python manage.py startapp catalog
```

```
cd ../mobile_service
django-admin startproject mobile_service .
python manage.py startapp catalog
```

```
cd ../staff_service
django-admin startproject staff_service .
python manage.py startapp portal
```

```
cd ../customer_service
django-admin startproject customer_service .
python manage.py startapp portal
```

2. Cấu trúc Docker Compose

```
docker compose config
```

```yaml
name: kiemtra01_03
services:
  api-gateway:
    depends_on:
      customer-service:
        condition: service_started
        required: true
      laptop-service:
        condition: service_started
        required: true
      mobile-service:
        condition: service_started
        required: true
      staff-service:
        condition: service_started
        required: true
    image: nginx:latest
    networks:
      app_network: null
    ports:
      - mode: ingress
        target: 80
        published: "8080"
        protocol: tcp
```

3. **Khởi chạy hệ thống**

```
docker compose up -d --build
```

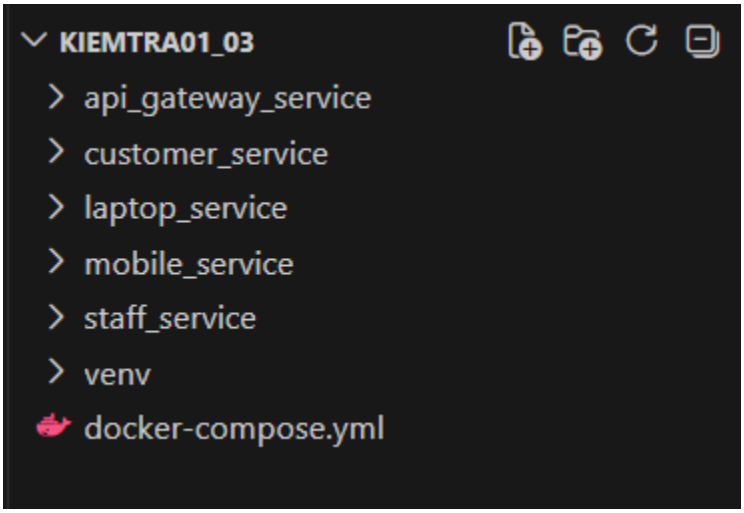
```
[+] up 23/29\CC\SAD\Kiemtra01_03> docker compose up -d --build
 - Image p... [⣿⣿⣿⣿⣿⣿⣿⣿⣶⣿⣿⣿⣿⣿⣿⣿⣿] 52.84MB / 133.9MB Pulling 42.6s
 - Image m... [⣶⣿⣶⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿] 99.62MB / 247.7MB    Pulling 42.6s
 ✓ Image kiemtra01_03-customer-service Built
 ✓ Image kiemtra01_03-mobile-service   Built
 ✓ Container kiemtra01_03-db-mobile-1  Running  0.0s
```

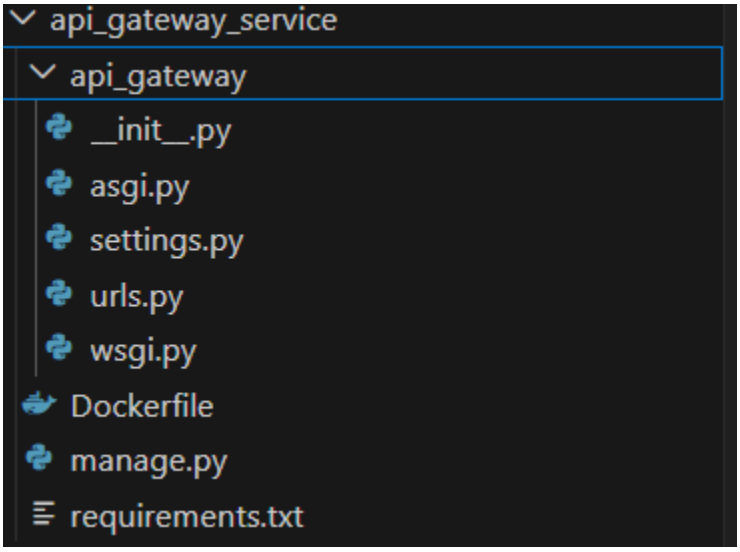
4. **Kiểm tra Logs của Microservice**

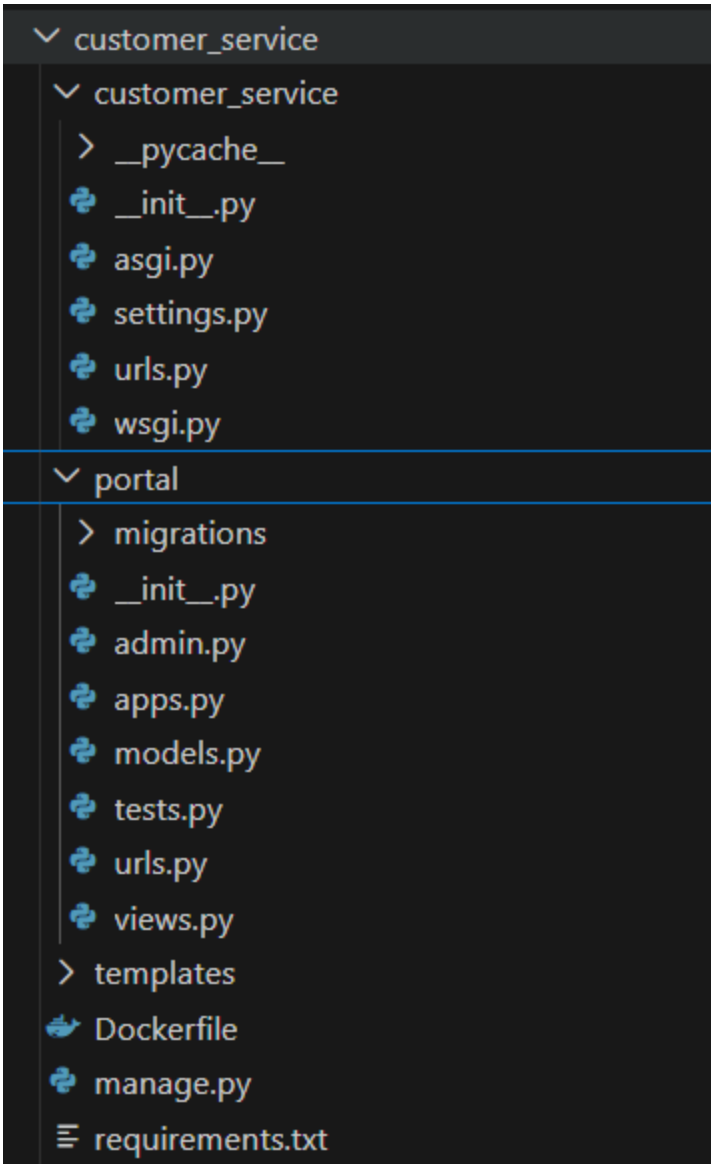
```
docker compose logs --tail 50
```

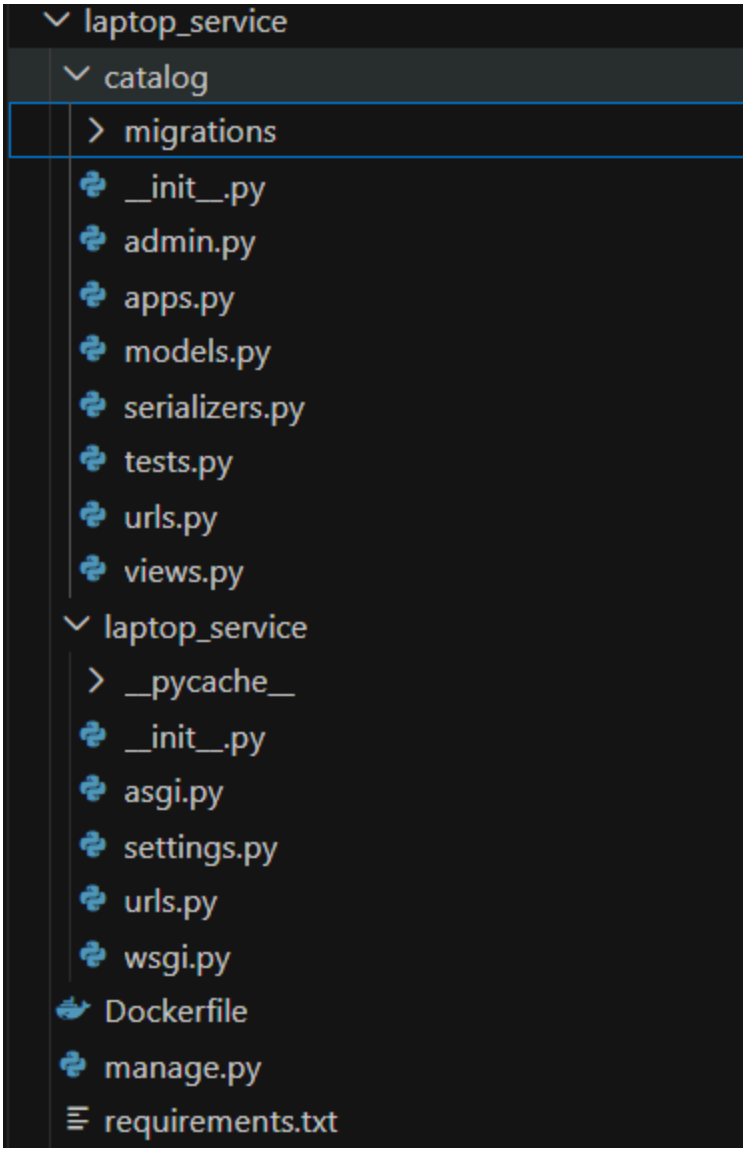
```
staff-service-1 | 2026-03-24 08:35:12 INFO [catalog] Requesting /api/laptops/
laptop-service-1| 2026-03-24 08:35:12 INFO [django.server] "GET /api/laptops/ HTTP/1.1"
200 458
api-gateway-1   | 172.18.0.1 - - [24/Mar/2026:08:45:00 +0000] "GET /staff/ HTTP/1.1" 200
1354 "-" "Mozilla/5.0"
```

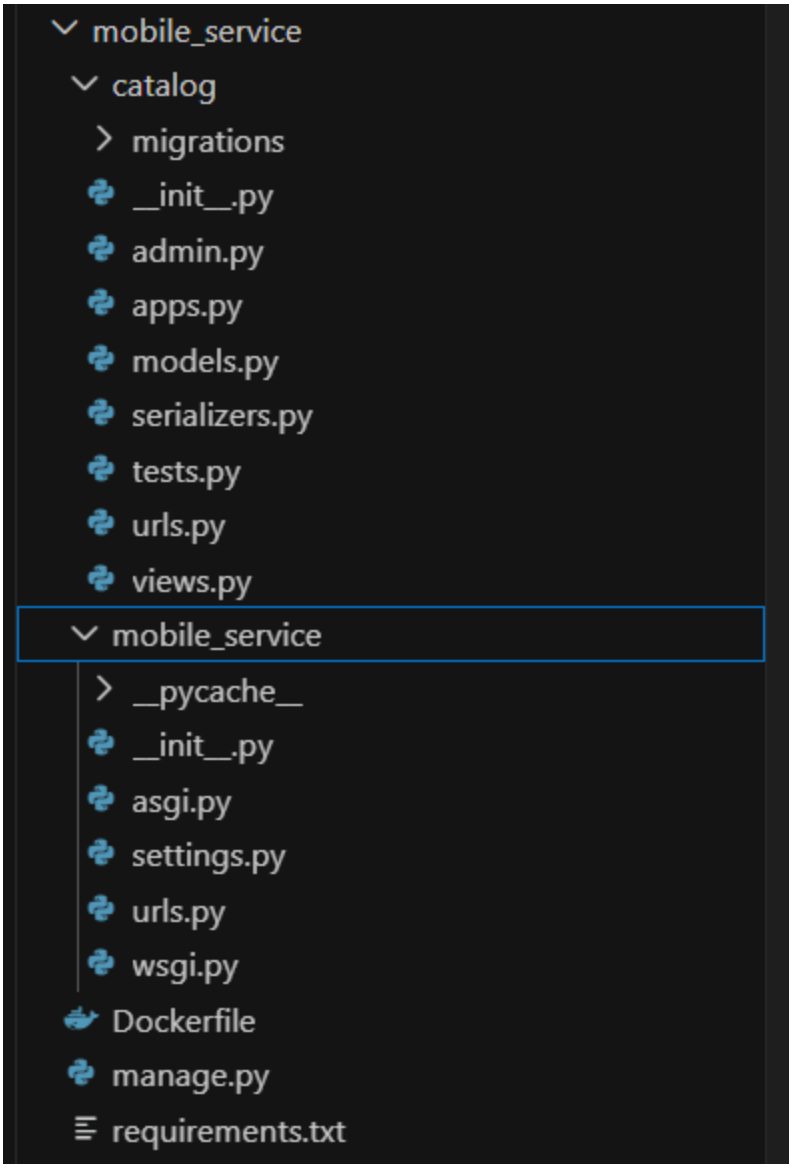
5. Cấu trúc file

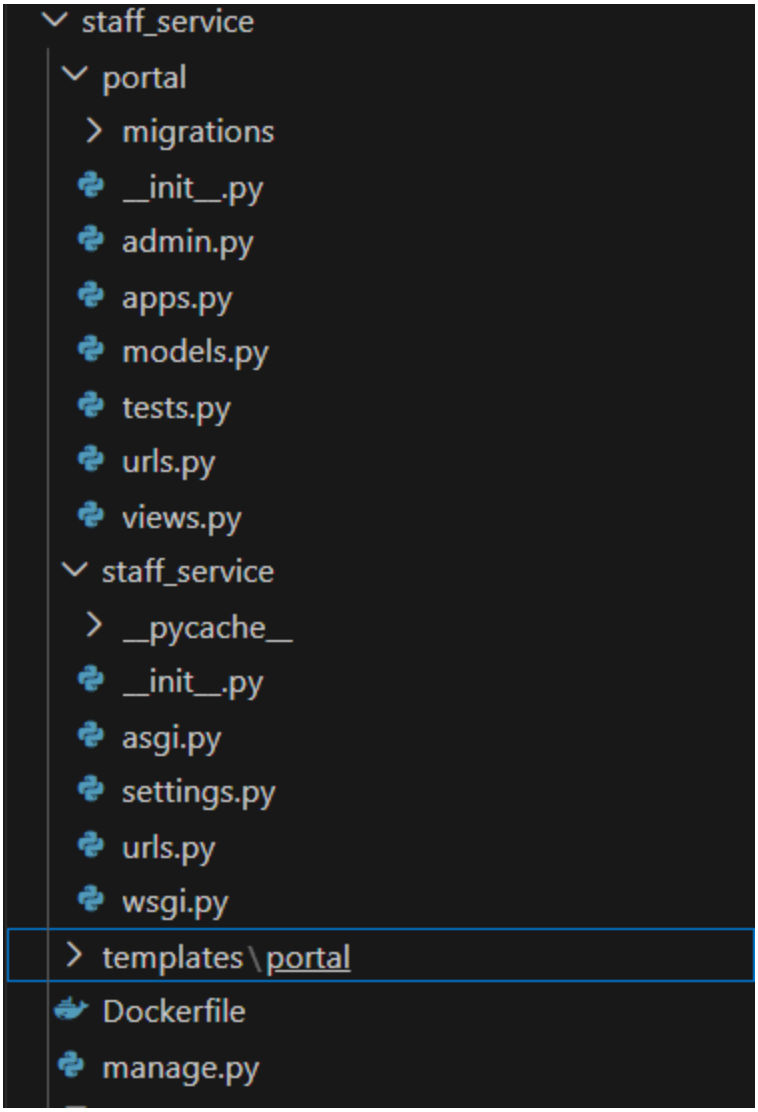





customer_service
customer_service
__pycache__
__init__.py
asgi.py
settings.py
urls.py
wsgi.py
portal
migrations
__init__.py
admin.py
apps.py
models.py
tests.py
urls.py
views.py
templates
Dockerfile
manage.py
requirements.txt

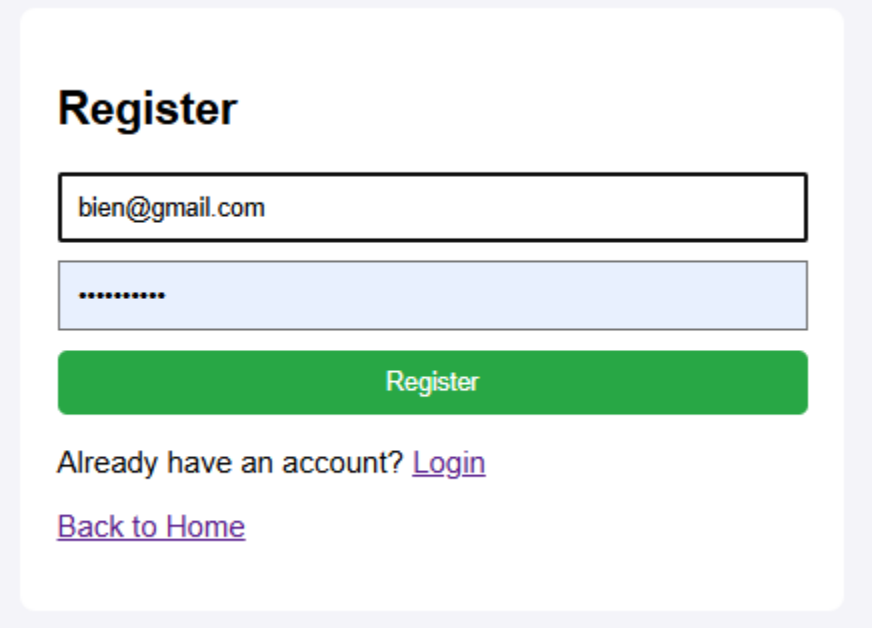

laptop_service
catalog
migrations
__init__.py
admin.py
apps.py
models.py
serializers.py
tests.py
urls.py
views.py
laptop_service
__pycache__
__init__.py
asgi.py
settings.py
urls.py
wsgi.py
Dockerfile
manage.py
requirements.txt


mobile_service
catalog
migrations
__init__.py
admin.py
apps.py
models.py
serializers.py
tests.py
urls.py
views.py
mobile_service
__pycache__
__init__.py
asgi.py
settings.py
urls.py
wsgi.py
Dockerfile
manage.py
requirements.txt

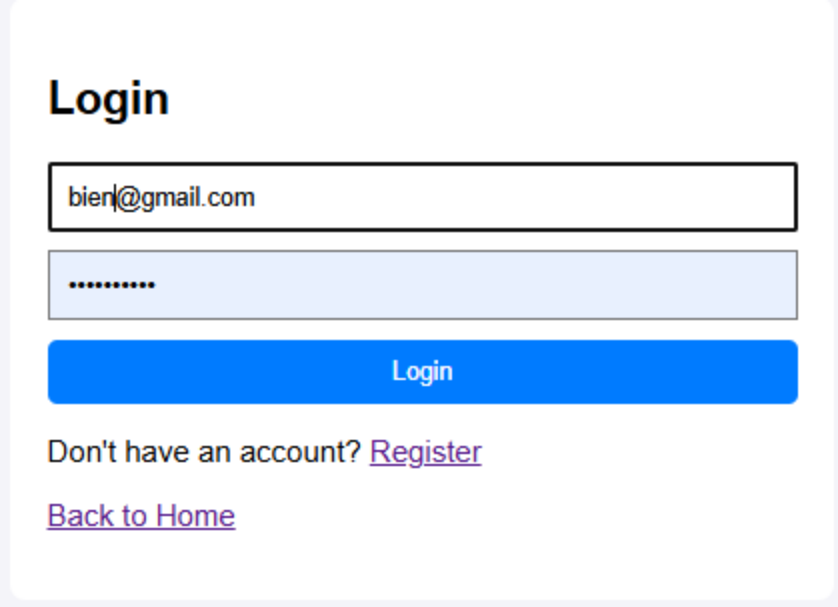


II. Giao diện kết quả

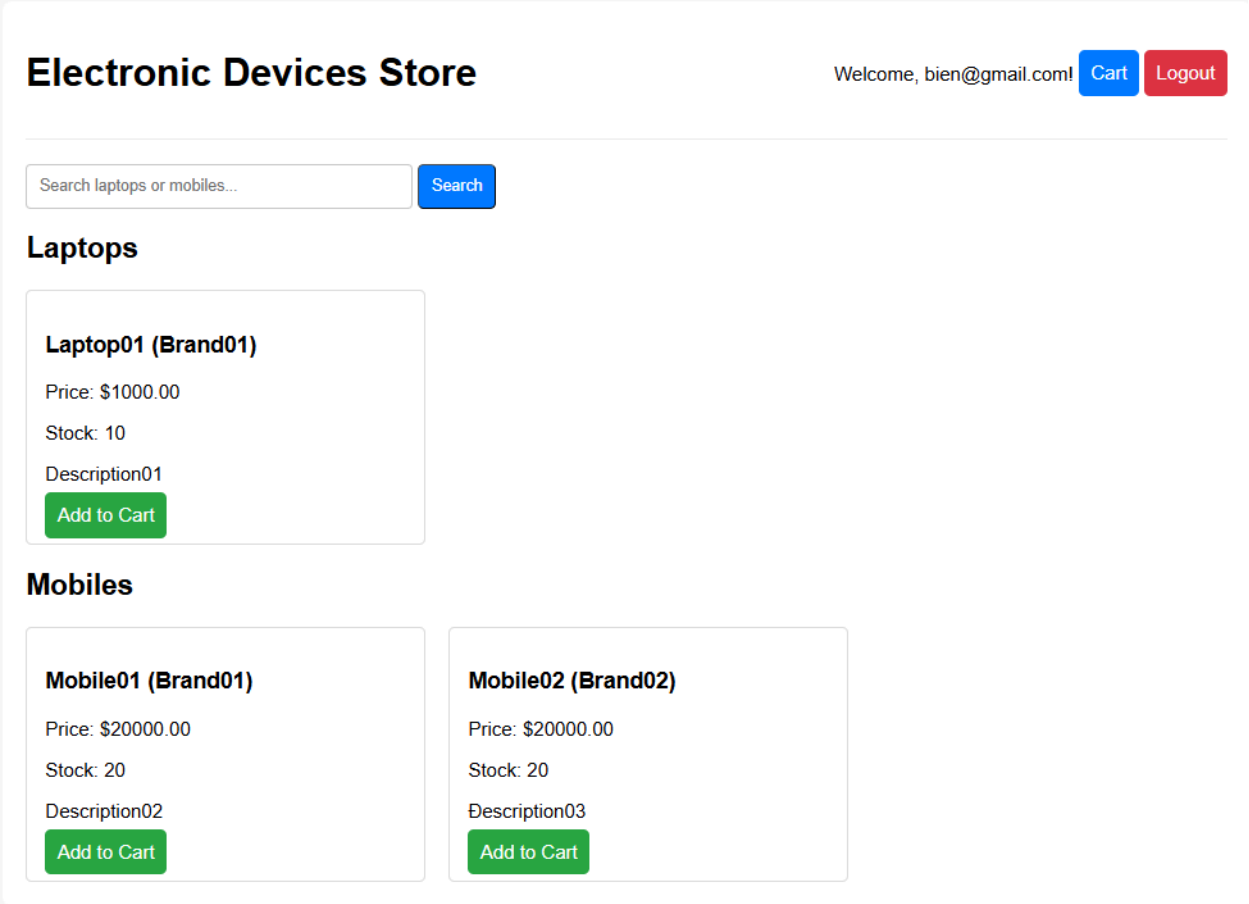
1. Đăng ký



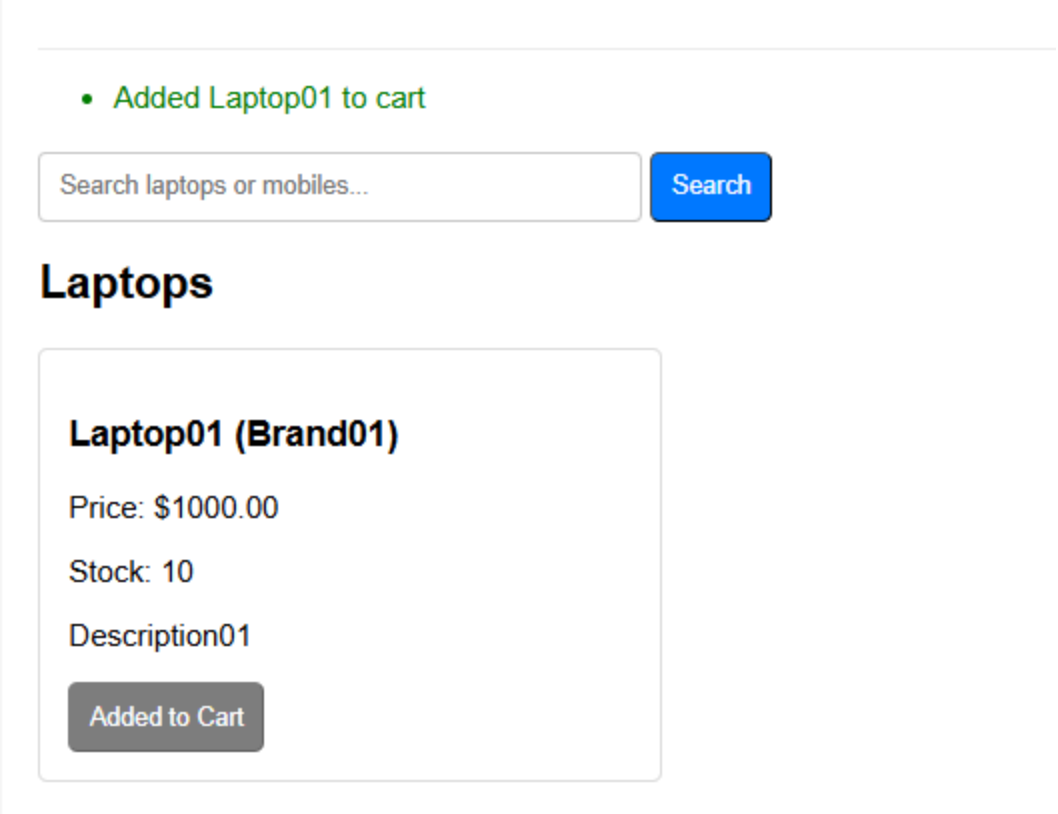
2. Đăng nhập



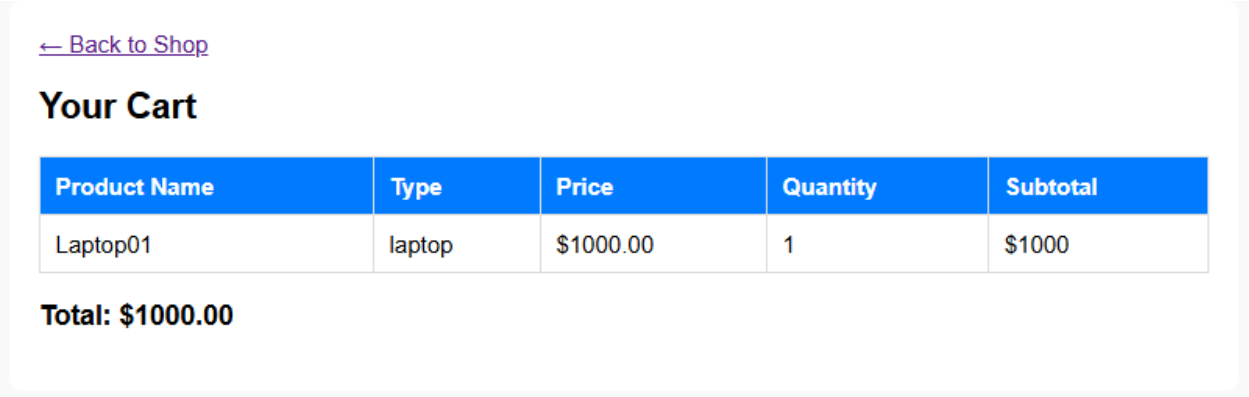
3. **Giao diện khách hàng**



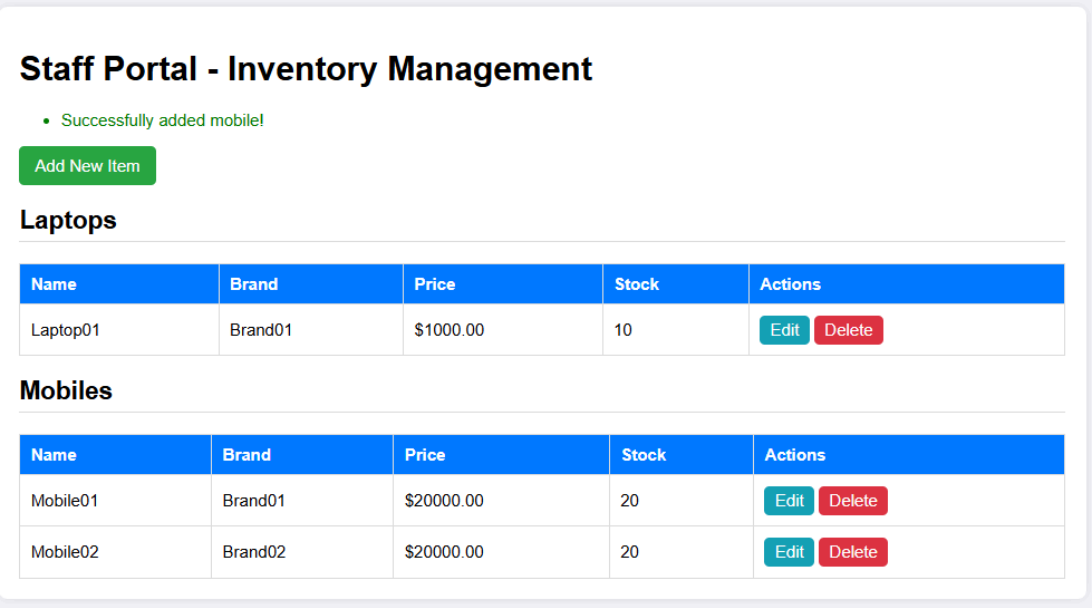
4. **Khách hàng thêm sản phẩm vào giỏ hàng**



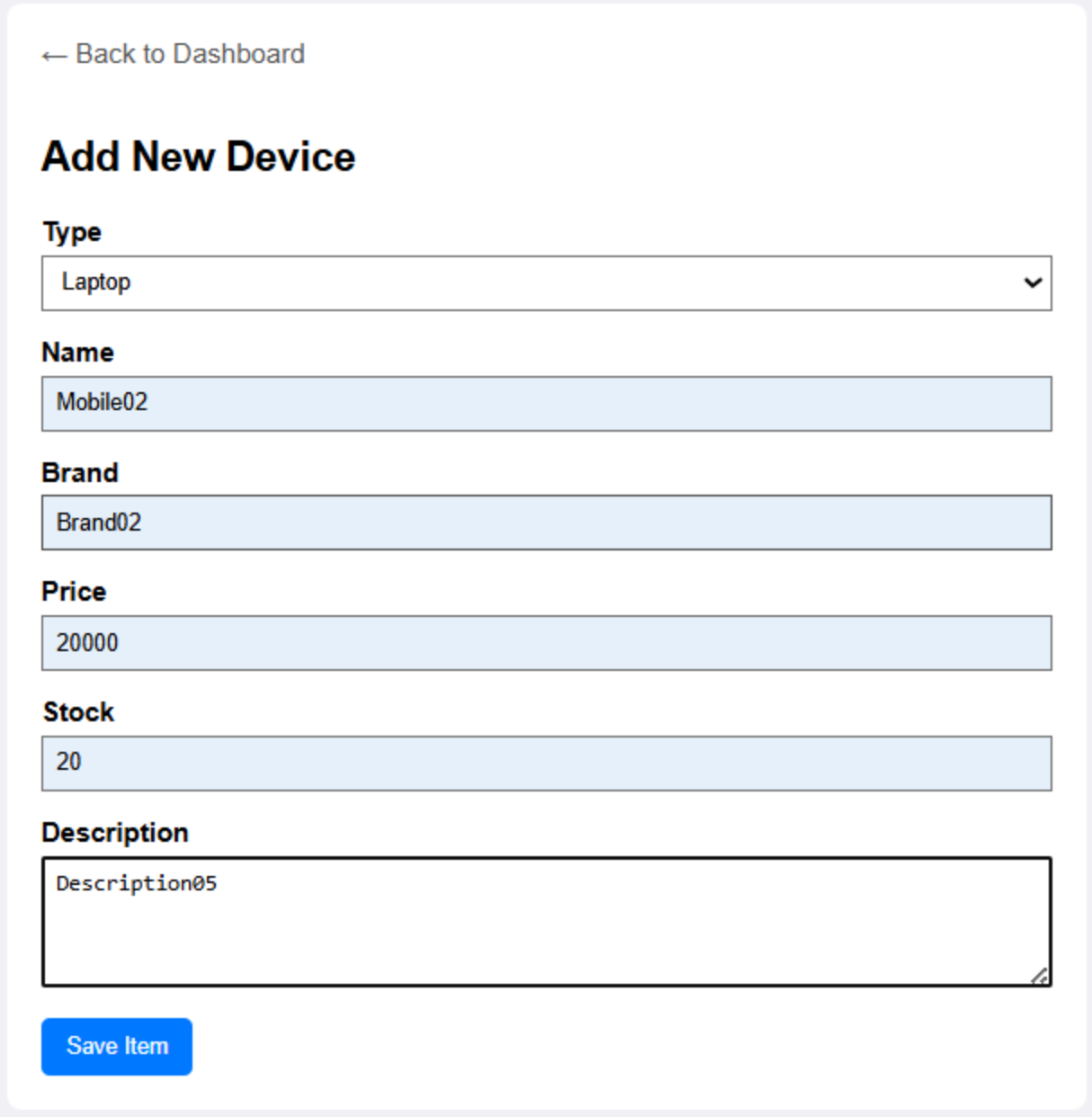
5. **Giỏ hàng**



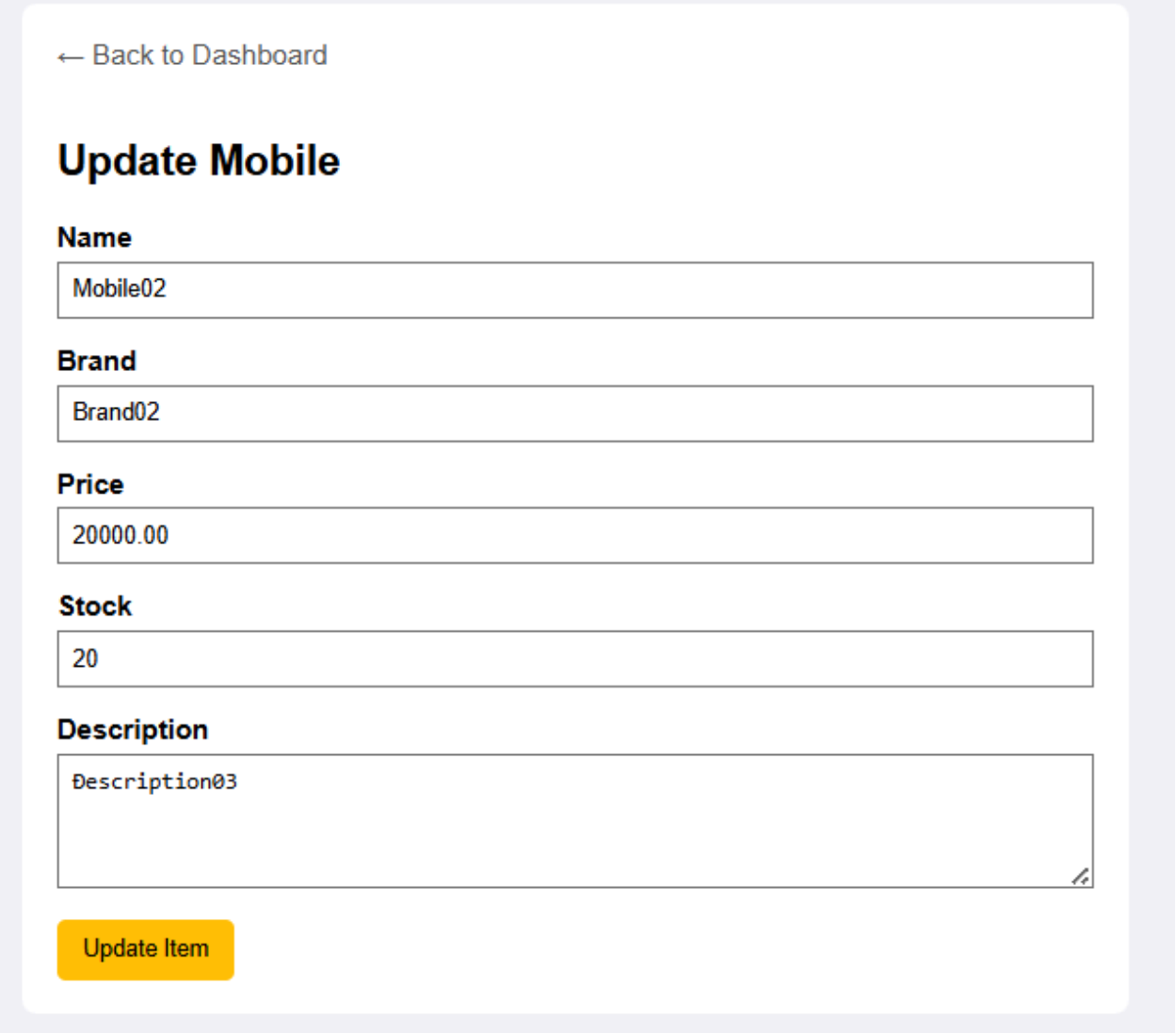
6. **Giao diện nhân viên**



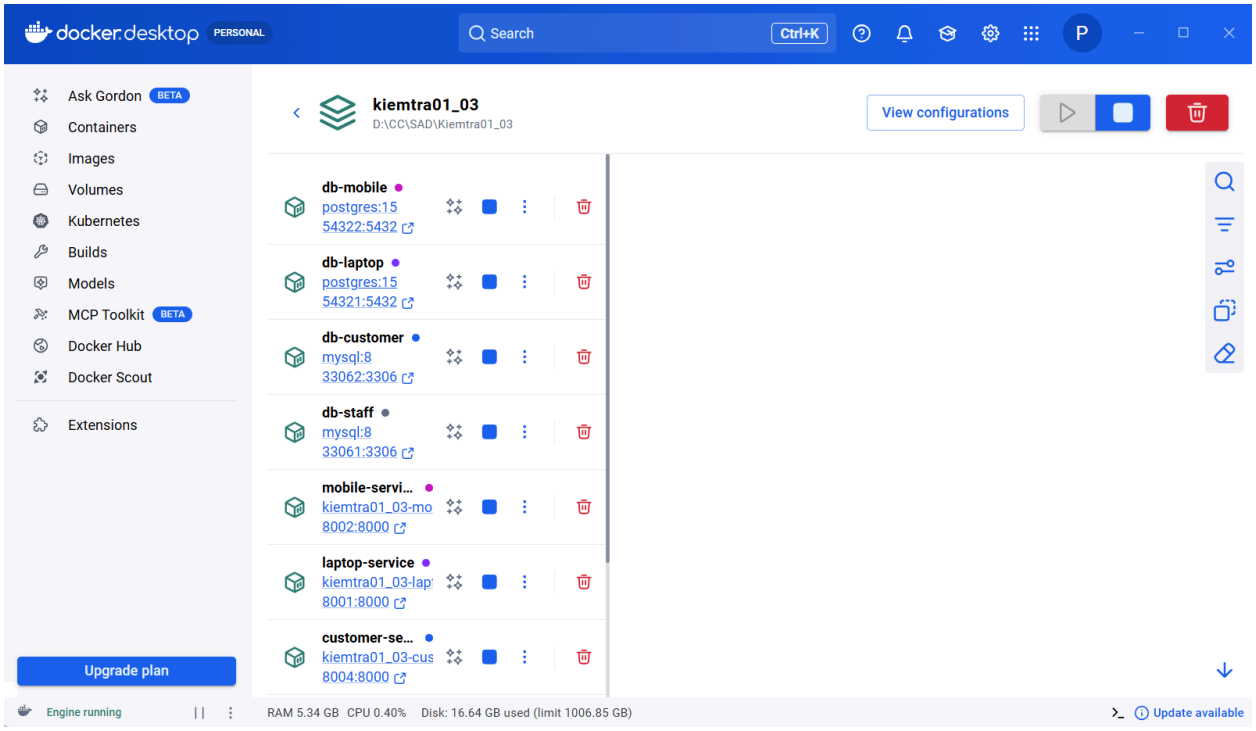
7. **Thêm sản phẩm**

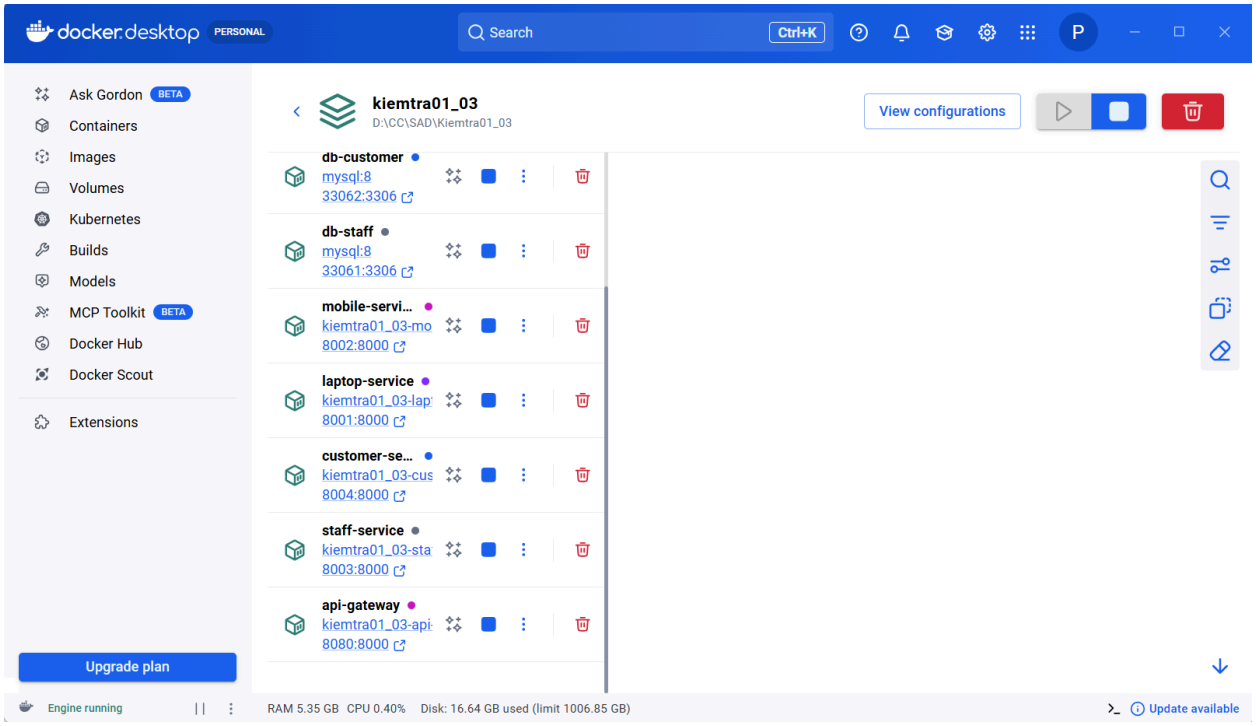


8. **Cập nhật sản phẩm**


← Back to Dashboard
Update Mobile
Name
Mobile02
Brand
Brand02
Price
20000.00
Stock
20
Description
Description03
Update Item

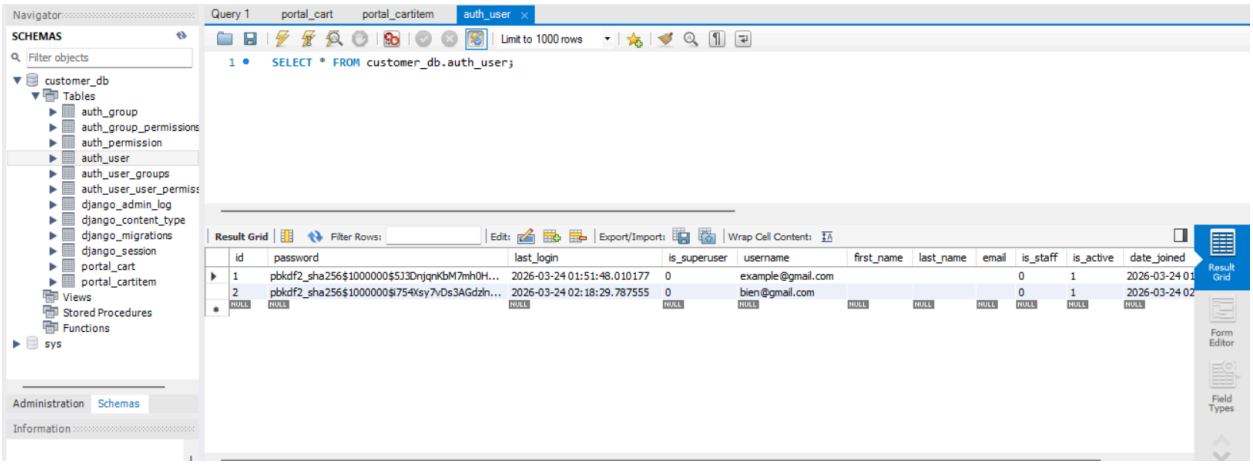
III. Docker


docker desktop
PERSONAL
Search
Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Engine running
kiemtra01_03
D:\CC\SAD\Kiemtra01_03
View configurations
db-mobile
postgres:15
54322:5432
db-laptop
postgres:15
54321:5432
db-customer
mysql:8
33062:3306
db-staff
mysql:8
33061:3306
mobile-servi...
kiemtra01_03-mo
8002:8000
laptop-service
kiemtra01_03-lap
8001:8000
customer-se...
kiemtra01_03-cus
8004:8000
RAM 5.34 GB CPU 0.40% Disk: 16.64 GB used (limit 1006.85 GB)
Update available

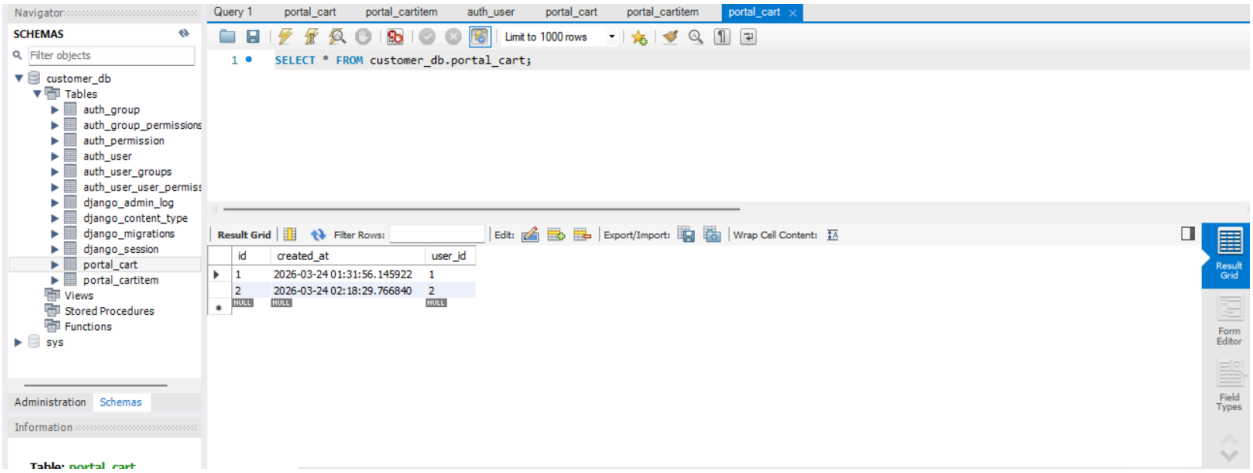


IV. Database

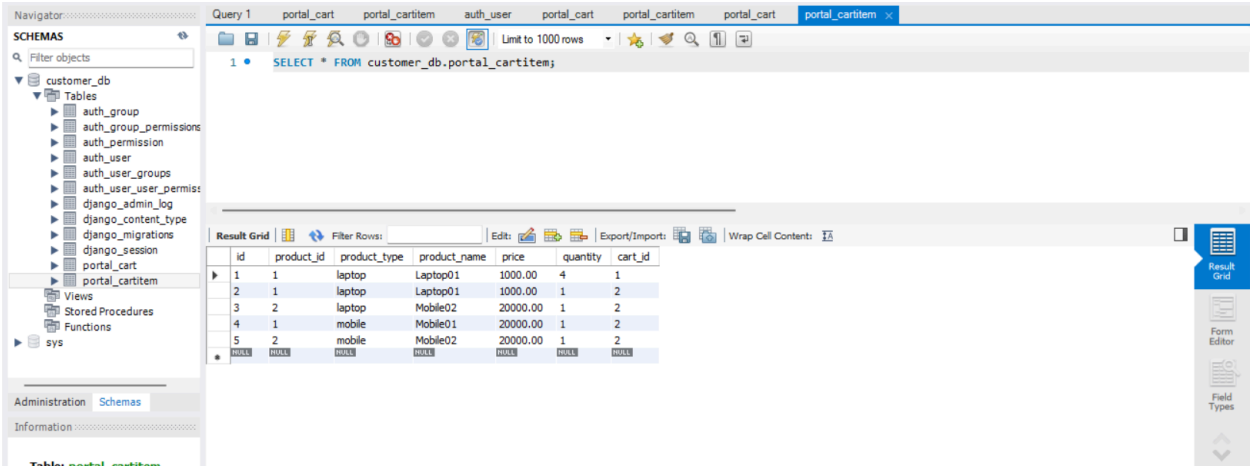
1. Db Customer



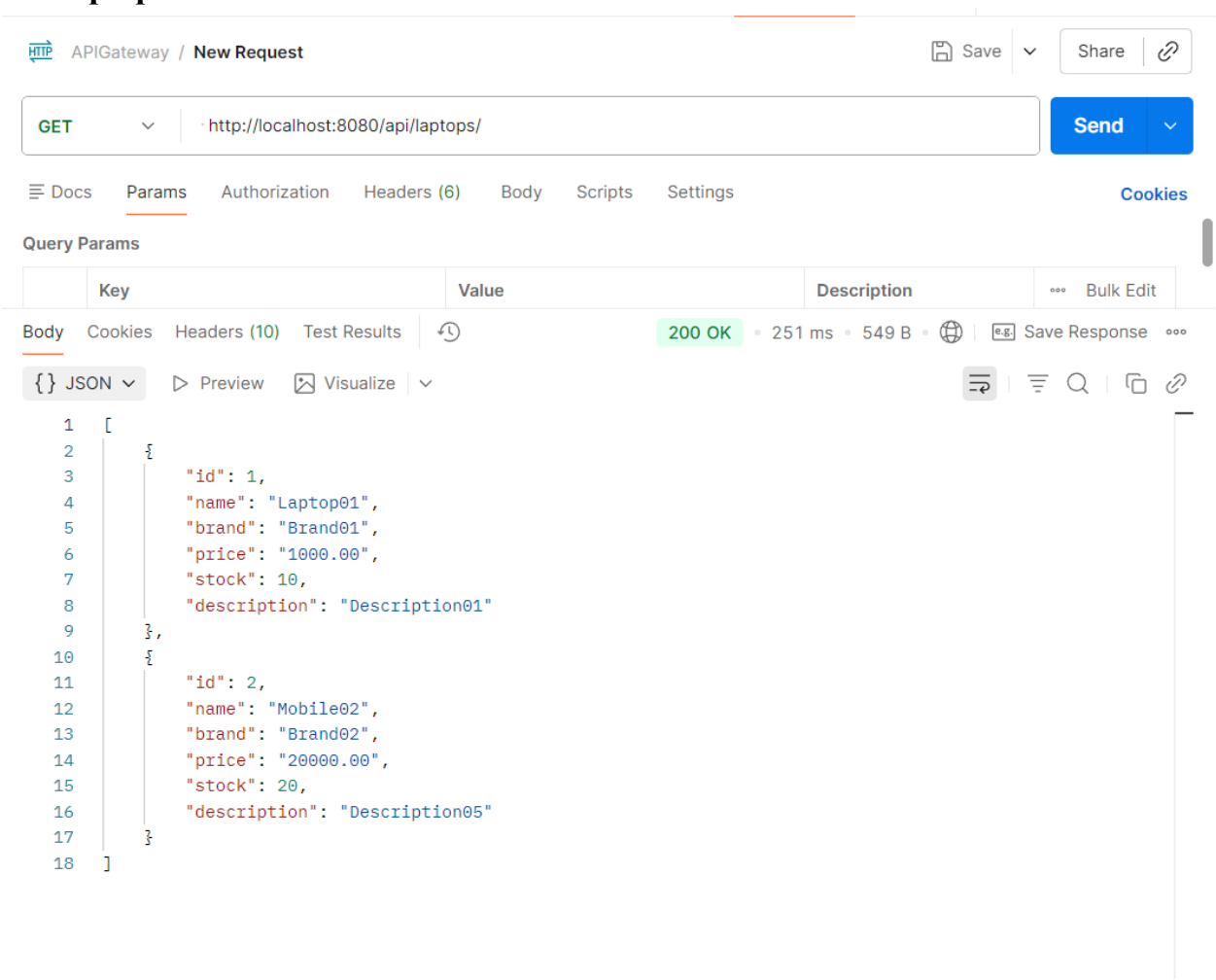
2. Db Cart



3. Db CartItem



4. Db Laptop



5. Db Mobile

GET http://localhost:8080/api/mobiles/ Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit

Body Cookies Headers (10) Test Results 200 OK 210 ms 551 B Save Response

{} JSON Preview Visualize


```
[
    {
        "id": 1,
        "name": "Mobile01",
        "brand": "Brand01",
        "price": "20000.00",
        "stock": 20,
        "description": "Description02"
    },
    {
        "id": 2,
        "name": "Mobile02",
        "brand": "Brand02",
        "price": "20000.00",
        "stock": 20,
        "description": "Description03"
    }
]
```